

Lab 03 - Research Critique 2

Research Critique 2

Read & annotate the research article

Instructions

First, do an initial read through of the article to get an overall sense of what was done and get an initial impression of some of the major issues that might be present.

Next, read through it again, slowly and methodically, to get a better sense of what they are doing in this experiment; think in terms of validity and confounds as you go. Make sure to annotate and take notes.

After reading and annotating the article, continue on to answer the questions.

Visual-Perceptual Speed of Karate Practitioners at Three Levels of Skill¹

Perceptual and Motor Skills, 1998, vol.87, pp. 96-98

Abstract: The Identical Pictures Test was administered to 50 male and 45 female volunteer karateka who were classified by the karate belt-rank color system (kyu-dan) into three groups. Separate analysis of variance (ANOVA) tests for each gender gave significant differences for visual-perceptual skill. Overall, earning a black belt leads to faster visual-perceptual speed.

Introduction. In many sports perceptual speed is important for a performer who needs to make a quick decision (Meese, 1981; Miller, 1960). In karate it is necessary to observe the opponent's direction, speed, and position to make rapid decisions regarding attack and defense. Research on visual-perceptual speed and stages of learning in sport is limited so differences in visual-perceptual speed of male and female karateka (karate practitioners) in three stages of learning was examined. It was hypothesized that longer study of karate would lead to faster visual-perceptual speed. Specifically, that holders of each successive belt would have faster visual-perceptual speed because of increases in the amount of time spent training.

Method. Fifty men (mean age = 23.1 yr.) and 45 women (mean age = 22.6 yr.), from the University Shotokan karate classes, ($ns = 20$ men, 15 women), the Shotokan karate club at the University ($ns = 15$ men, 15 women), or a local Shotokan Karate dojo (training hall) ($ns = 15$ men, 15 women) volunteered. The classes and club were taught by the same person, and the dojo was selected because of its similarity with the university classes and club, although a different instructor taught at the dojo. These subjects were divided into three groups based on their current belt-rank: White belts ($ns = 20$ men, 15 women), blue belts ($ns = 15$ men, 15 women), and the black belts ($ns = 15$ men, 15 women). At the time of grouping the subjects, the white belt group had practiced karate for less than 12 months, the blue belt group had practiced from 8 to 18 months, and the black belt group was comprised of those who had practiced karate for more than 18 months. All three groups trained for an additional 4 weeks before their visual-perceptual speed was tested. Sensei Kim instructed the black belt group, while one of Sensei Kim's blue belt instructors taught both the white and blue belt groups.

¹The original article has been modified for the purposes of this assignment.

The Identical Pictures Test is a paper-and-pencil test designed to measure visual-perceptual speed and was selected because it has been reported as reliable (Stallings, 1968), administration is easy, and it has been used in research on other sports (Meese, 1981; Miller, 1960). The test has two parts of 48 items, each of which are timed for 1.5 min. Each correct item is scored 1 point, so a perfect score is 96 points. The higher the score, the faster the visual-perceptual speed. To standardize instructions, a prerecorded audiocassette was played to explain the purpose of the study, the instructions for the test, and the timing for the two parts of the test.

Results and Discussion. The mean scores and standard deviations by skill levels for males and

Table 1.

Means and Standard Deviations for Identical Pictures Test

Belt Rank									

females are presented in Table 1.

Analyses of variance were calculated to examine the significance of differences of skill for women ($F(2,47) = 4.17, p < .05$) and men ($F(2,27) = 10.43, p < .05$). The Tukey *post hoc* test indicated the same pattern for both men and women; those with black belts had faster visual-perceptual speed than those with blue belts or white belts, but there was no difference between blue belts and white belts.

The present results suggest that acquiring expertise in Shotokan, Rei-Shobu-Kan karate or other martial arts will improve performance in sporting activities in which people must make decisions quickly, or those that require fast perceptual speed for successful performance. In addition, these results suggest that initial improvements in karate skills (i.e., those acquired during advancement from white to blue belt) don't translate to improved visual-perceptual skills. Rather it is only the acquisition of advanced karate skills (i.e., skills typical of a black belt) that produces these increased visual-perceptual skills.

The reader may want to consider that the black belt group's higher scores may reflect improvement that has resulted from their years of martial arts training, or alternatively, that of those who begin karate training, only those with high visual-perceptual skills achieve the rank of black belt.

Evaluate & critique (70 points)

Instructions

Part A. Identify the research design (36 points; 2 each)

Answer the following questions based on the research you read. Remember that you can (and should!!) use the handouts and other course materials to guide you in this process.

1. Hypothesis:

a. Do the researchers intend to investigate an associative or a causal relationship?

Answer:

- b. What language did they use that told you which type of RH they intended? *Remember, this is a question about what they want to do, not what they actually did!!*

Answer:

2. **Internal Validity:** If the researchers were interested in a causal interpretation of the results, then you must consider the procedures used in the study, and their resulting impact upon internal validity. Describe the following aspects of the study in detail, being sure to use the proper terminology that you've learned in class.

- a. What was the method of participant assignment?

Answer:

- b. What was the IV, and how was it manipulated?

Answer:

- c. Was ongoing equivalence maintained? If not, what issues did you note? (Be very careful here. If they did not provide enough info to make a determination about ongoing equivalence, then you should not assume there is any.)

Answer:

- d. Regarding ongoing equivalence: Given the intended research hypothesis, can you think of any serious confounds for ongoing equivalence? Give an example below.

Answer:

- e. Given your answers to a–d, is there causal interpretability of the results, why or why not? Does this fit with your answer to question #1a, and if not, what does this mean?

Answer:

3. **Measurement Validity:** Think carefully about each concept is measured in this study.

- a. What is the IV labeled as? (how did they describe the IV in terms of the RH)

Answer:

b. How is the IV operationalized?

Answer:

c. Does the IV as it is labeled seem to “agree” with how it is operationalized? In other words, are they actually measuring the concept they *intended* to measure here?

Answer:

d. What is the DV?

Answer:

e. Is the DV they used an adequate measure of the construct that they intended to measure (refer to the research hypothesis)? Why or why not?

Answer:

4. **External validity:** External validity is a potential concern with every study... ever, because we must decide how broadly or narrowly we can expect the results from a study to generalize.

a. What is the population being studied? (who do they **want** to generalize to)

Answer:

b. How would you describe the sample that they obtained? Be sure to provide information about both the sampling method and the resulting data sample.

Answer:

c. What is the setting for the procedure? What is the setting for the task that obtained a DV? (Again, if it was not stated, do not make assumptions!)

Answer:

4. Generalizability

- a. What generalization did the researchers make? (Be careful here, remind yourself of what a generalization is first, we're not talking about the results in this study, but what those results can extend to!)

Answer:

- b. What is your evaluation of their generalization? Is the generalization too narrow, too broad, or appropriate?

Answer:

- c. Let's apply these findings to another context. Since you are knowledgeable about this study, your friend asks,

"I've been trying to decide if I should I have my 10-year-old take tennis lessons while she is at camp next summer to improve her perceptual-motor and interpersonal problem-solving skills? This study seems to suggest it is a good idea, since it found that improvements in physical skills corresponds with improvements in visual-perceptual and decision-making skills."

Keep in mind!

Your overall answer should also take into consideration how the grouping/independent variable was operationalized. You should also note the details of why your friend should, or should not, apply the results of the current study to their situation. (Think in terms of each of the details in which a generalization is being made).

How would you answer your friend?:

Instructions**Part B. Evaluating alternate research hypotheses (44 points; various)**

Let's reconsider the final line in the karate article (the parenthetical numbers have been added)...

"The reader may want to consider that (1) the black belt group's higher scores may reflect improvement that has resulted from their martial arts training or, alternatively, that (2) of those who begin karate training only those high visual-perceptual skills achieve the rank of black belt."

The authors suggest that there are two possible explanations of the results, that 1) Training in karate led to higher visual-perceptual skills, as evidenced by the black belt scores on the DV, or 2) Training in karate had nothing to do with increasing the DV scores, and in fact, no one's visual-perceptual skills changed, it's just that some of them already had high visual-perceptual skills, and it's this preexisting skill that allowed some people to achieve the rank of black belt.

We want to practice redesigning the original study to provide a better test of these alternative research hypotheses. In the next few questions, you will need to describe what we need to do to test the specified RH.

1. RH#1: Longer study of karate training causes improvements in visual-perceptual skills.

Note on what is plausible, and ethical, in the conduct of research

When we say plausible, we mean is it realistic to be able to do. For example, we could control internal validity/ongoing equivalence by monitoring participants 24/7; however, it's not plausible because no one would submit to that, and it would take an enormous research staff to pull it off. It just simply isn't realistic.

When we say ethical, we mean that just because we could answer a research question with a given design, doesn't make it okay to run that design. Think back to the famous Milgram experiment or the Stanford prison experiment; you likely learned about both in your Intro Psych class. We could never run them today because they are unethical to put participants through.

In terms of the karate study, it would be unethical to randomly assign participants to a karate class condition since there is a risk of injury involved (we could only use volunteers who self-select into the karate condition, or those already in karate).

Before you go further, I'd like you to take a moment and reflect on how the last few sentences impact the requirements for a true experiment!!!

a. (10 points). Between Groups Experimental Design for RH#1.

The Karate study you read attempted to test a causal hypothesis (RH#1) but did not meet the criteria for a true between groups experiment. In the space below, identify and take a screenshot Windows-Shift-S (windows), Shift-Command-4 (mac) of the slide from the Internal Validity lecture that lists the "Acceptable assignment procedure for Causal RH." Paste that screenshot here below.

Lecture slide screenshot:

Next, review the content on the lecture slide you identified, and describe how you could **redesign the participant assignment phase of this study** to test RH#1 between subjects. In your answer, make sure to explain how you plan to accomplish the method of participant assignment as well as how you plan to manipulate the IV.

Note: For the purposes for this question, assume that you are NOT limited by any ethical or practical limitations. Also, focus specifically on the participant assignment phase and do not worry about ongoing equivalence (for the moment).

Research design:

b. (8 points). Between Groups Non-experimental Design for RH#1

Now, imagine that the constraints of ethics and practical reality have been “turned back on” and you are left trying to test RH#1 with a non-experimental design. Your method of participant assignment and IV must remain the same as the original Karate study but you have the opportunity to redesign how the procedures will be standardized across belt-groups to better maintain ongoing equivalence.

In the space below, identify and take a screenshot Windows-Shift-S (windows), Shift-Command-4 (mac) of the slide from the Internal Validity lecture that lists the “2 things to “look for” to help you evaluate ongoing equivalence.” Paste that screenshot here below.

Lecture slide screenshot:

Next, review the content on the lecture slide you identified, and describe how you could **redesign the procedural standardization phase of the study** to better maintain ongoing equivalence.

Answer:

c. (2 points). Consider your answer to the previous question in (b).

If the researchers were able to implement all of your suggested changes, would this new study provide an acceptable test of RH#1? Why or why not?

Answer:

d. (6 points). Within Groups Design for RH#1.

So far, we’ve been thinking in terms of comparing effects between groups; however, RH#1 is essentially asking if peoples’ visual-perceptual skills change over time as a function of their karate training. Wouldn’t this make the most sense as a within-groups design? Is it possible to have a within-groups, true experimental test of this RH? Why or why not?

Answer:

- e. **(4 points).** What about just using a within-groups, non-experiment, to test RH#1? We could just measure each person's visual-perceptual speed when they join the karate class, and then measure it again after karate training, and attribute any difference between the two tests to the karate training. What's wrong with that?

Answer:

2. RH#2: Of those who begin karate training, only those with good visual-perceptual skills will achieve the rank of black belt.

Keep in mind!

This RH might be restated as... "Practicing karate doesn't improve visual-perceptual skills, rather only those who have good visual-perceptual skills when they begin the study of karate stick with it (eventually becoming black belts)."

- a) **(4 points).** Can you devise a true experiment (BG or WG) to test RH#2? Why or why not?

Answer:

- b) **(10 points).** How *would* you go about testing this RH with a non-experimental (BG or WG) design?

Keep in mind!

Because we are asking you to craft a non-experimental design, this new study will not provide a causal test of RH#2, but you should still make informed decisions about the design to minimize confounds. For this question, you should design a study that conforms to the constraints of real world ethics and practical reality (within limits). Make sure to include a clear description of the sampling and data collection procedures, as well as the IV, DV, and any control variables in your new study design. It should also be clear whether your design is attempting to examine this relationship between groups or within groups.

Hint: Go back to review the handouts and the lecture slides to help you with this question!!

Research design: